Satellite Video Scene Classification Using Low-Rank Sparse Representation Two-Stream Networks

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***Abstract*— Satellite video scene classification (SVSC) is a challenging work in remote sensing. The main procedure of SVSC is spatial–temporal feature extraction. Unfortunately, massive numbers of dim small moving targets and the low signal-to-noise ratio (SNR) of satellite video bring great challenges to feature extraction. It is difficult to apply traditional feature extraction methods to SVSC because they are used to classify the actions of high-quality video. According to the theory of low-rank sparse decomposition, a Low-rank Sparse Representation Two-stream Network (LSRTN) is designed to increase the classification accuracy of two-stream networks. First, we propose a Low-rank Sparse Component Analysis Network (LSCAN) to decompose satellite videos into low-rank background images and sparse moving target sequences. The LSCAN possesses the advantage of low-rank sparse decomposition to solve small targets and has the capability to adjust the features using the data. Moreover, the LSCAN can efficiently improve the feature extraction of low SNR video. Second, a two-stream structure that was proven to be effective for multiclass video classification was applied to obtain the spatial features and temporal features in each stream. Finally, a fully connected layer integrates the features to classify the satellite video scenes. To utilize the label information, we refine the loss function to adjust the degree of low-rank sparse characteristics and ensure the classification accuracy of training. The experimental results demonstrate that the proposed method achieves better performance than the baseline methods for the SVSC task.**

***Index Terms*— Satellite video, scene classification, two-stream networks, video classification.**

I. Introduction

With the rapid development of satellite remote sensing techniques, the acquisition frequency of satellite video has become almost real-time. Compared to multitemporal satellite image sequences, satellite video demonstrates the potential ability to capture dynamic information in remote sensing scenes. The observation interval of satellite video is shorter than that of multitemporal satellite image sequences, which gives satellite video the ability to capture more spatial–temporal information.

It is essential to perform more studies on the characteristics of satellite videos. Recently, some target detection and tracking studies have focused on the moving targets of satellite videos. Those studies gave traditional applications of videos but may ignore the characteristics of remote sensing. More potential applications based on remote sensing characteristics should be developed. The topic extends from the video classification in common video understanding and scene classification in the remote sensing image processing area.

The difference between video understanding in computer vision and satellite-based remote sensing is that satellite video can capture a huge amount of observation information in a large-scale scene. Although the observation scale is large, only a few regions contain massive numbers of moving targets, and the background is almost unchanged. In addition, existing satellite videos only last a short time due to the limitation of the satellite networking technique.

Video understanding is a hot topic that is useful for data analysis and creating an information database [1], [2]. This can be a direction to support more significant applications of satellite video. An example application is obtaining meaningful fragments from a massive amount of fragments to obtain a summary of the entire area in a period. It needs to attach semantic information to small areas according to the context. Satellite video scene classification (SVSC) is a key step to realize satellite scene understanding [3].

SVSC is defined as recognizing dynamic information in the observed scene with predefined dynamic scene types in a relatively small region due to properties of the satellite video, i.e., a few targets moving and an almost unchanged background. A satellite video with a large ground region can be composed of video blocks called satellite video scenes, and those scenes can cover specific events. SVSC aims to label those satellite video scenes. Similar to video classification in computer vision, the main challenge of SVSC is how to represent and extract spatial–temporal dynamic information effectively [4].

Benefiting from the wide range and real time monitoring capability, satellite videos open a new way to real-time understand dynamic scenes on the Earth. Satellite videos bring more

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information than the traditional multitemporal satellite image [5]–[7]. For example, satellite videos can monitor moving vehicles on a large scale of the region and across multiple regions, especially in some special regions where there are no navigation and communication signals. This case is really happened when a natural disaster occurs, such as landslide, earthquake, and flood. In those cases, satellite videos can help people to real-time monitor dynamic events.

Existing video classification methods have a certain reference value for spatial–temporal feature extraction. 3-D convolutional neural networks (3-D-CNNs) [8]–[11], CNN-based long short-term memory (CNN-LSTM) networks [12], [13], and two-stream networks are the main methods [14]–[16]. In theory, a 3-D-CNN can fully exploit both the spatial information and temporal information in satellite videos by considering all frames [17]. Furthermore, such a full network requires a huge number of training samples for better learning spatial–temporal features. Therefore, current studies on using 3-D-CNNs for video classification generally focus on designing and improving the structure of 3-D-CNNs. Different from 3-D-CNN, CNN-LSTM deals with spatial information and temporal information serially. A CNN-LSTM first extracts the spatial features from each frame and then extracts the temporal features from the sequence of spatial features to acquire the spatial–temporal features [18]–[20]. Compared to CNN-LSTMs, two-stream networks, as another typical approach to video classification, adopt a parallel framework in which spatial features are extracted from a single frame, and temporal features are extracted from either optical flow. Some methods based on two-stream networks have achieved better performance than 3-D-CNNs and CNN-LSTMs [15], [21]–[23]. In addition, integrating the spatial and temporal features of training and testing procedures is an ongoing topic for the deep learning methods mentioned above.

Due to the noticeable difference between computer vision and satellite remote sensing, researchers should pay more attention to extracting the spatial information of dim small targets and accurately expressing the dynamic information of moving targets. These two problems mentioned above are the challenges that SVSC has to overcome. Therefore, the existing 3-D-CNNs, CNN-LSTMs, and two-stream networks of computer vision cannot be directly used for SVSC. For example, a 3-D-CNN extracts the spatial–temporal features from all frames, but the motion of massive targets is unpredictable. Lacking training samples will become an unsolvable problem as the dim increases. Most objects, such as various cars, are small and weak in satellite video. The main difficulty that the CNN-LSTM has to address comes from dim small targets in satellite video when a CNN-LSTM is applied to conduct feature extraction for SVSC. It is difficult to capture the static features of targets from a single frame, which will cause the LSTM method to fail. Relatively speaking, two-stream network-based methods avoid the small target motion problem. In those methods, the spatial stream extracts global spatial features without considering the targets as the temporal stream extracts the motion information using the optical flow. Unfortunately, the optical flow may not function in some videos due to the strong noise, which leads to classification mistakes. Noise and jittering may drown the targets in one frame or some continuous frames, which usually does not occur in traditional video processing and severely influences optical flow extraction. Extracting optical flow [24], [25] is time-consuming, and it ignores the spatial features of moving targets.

A remarkable feature of satellite video is that the background of the observed scene is almost unchanged as the targets occupy only a few pixels. Satellite video has relatively dim and small but massive targets; we mainly focus on extracting the motion features of dim small targets. Namely, the background is of low rank, and the targets are sparse [26]–[30]. Therefore, separating satellite video scenes into low-rank parts and sparse representation parts could enhance the feature extraction ability. Fig. 1 shows the low-rank and sparse representation of a satellite video.

In the proposed method, we want to enhance the input of the two streams in two ways. The first way is to improve the feature extraction and representation ability. The other is to mine the relations contained in the data instead of collaborative features. Prior knowledge of low-rank sparse representation can improve the stability and accuracy of the model. In this article, the Low-rank Sparse Representation Two-stream Network (LSRTN) is proposed to learn spatial–temporal features from satellite videos for scene classification. First, by consulting the low-rank sparse composition, we design a learnable net to obtain an efficient and pertinent decomposition. Then, a ResNet [31] is applied to extract the spatial features from the low-rank part on one stream, and another ResNet extracts the temporal features of the other stream. Finally, the spatial and temporal features are fused by a fully connected layer to classify the videos [22]. In the training step, we refine the loss function to adjust the degree of low-rank sparse characteristics and ensure the training accuracy of classification to utilize the label information.

To evaluate the proposed method, SVSC experiments were conducted on a real satellite video dataset. In the datasets, classes of the satellite video include busy scenes and free scenes. Classification performance is used to evaluate the proposed method. The proposed method was compared to the state of the art. The experimental results demonstrated the effectiveness and efficiency of the proposed method. The static scenes are excluded from the dataset by the events of interest (EOIs) detection [32] method because dynamic scene classification is much more valuable in practical applications. State-of-the-art computer vision methods were compared with the proposed method on the dataset.

This work makes two contributions.

1) An Low-rank Sparse Component Analysis Network (LSCAN) is used instead of the traditional decomposition methods. An LSCAN is learnable and can properly address the data structure so that LSCAN can adjust the feature representation by combining the prior information and the labels.
2) A two-stream convolutional network is applied to achieve more accurate multilabel classification using spatial–temporal information. Prior structured features are helpful to increase the classification accuracy, and

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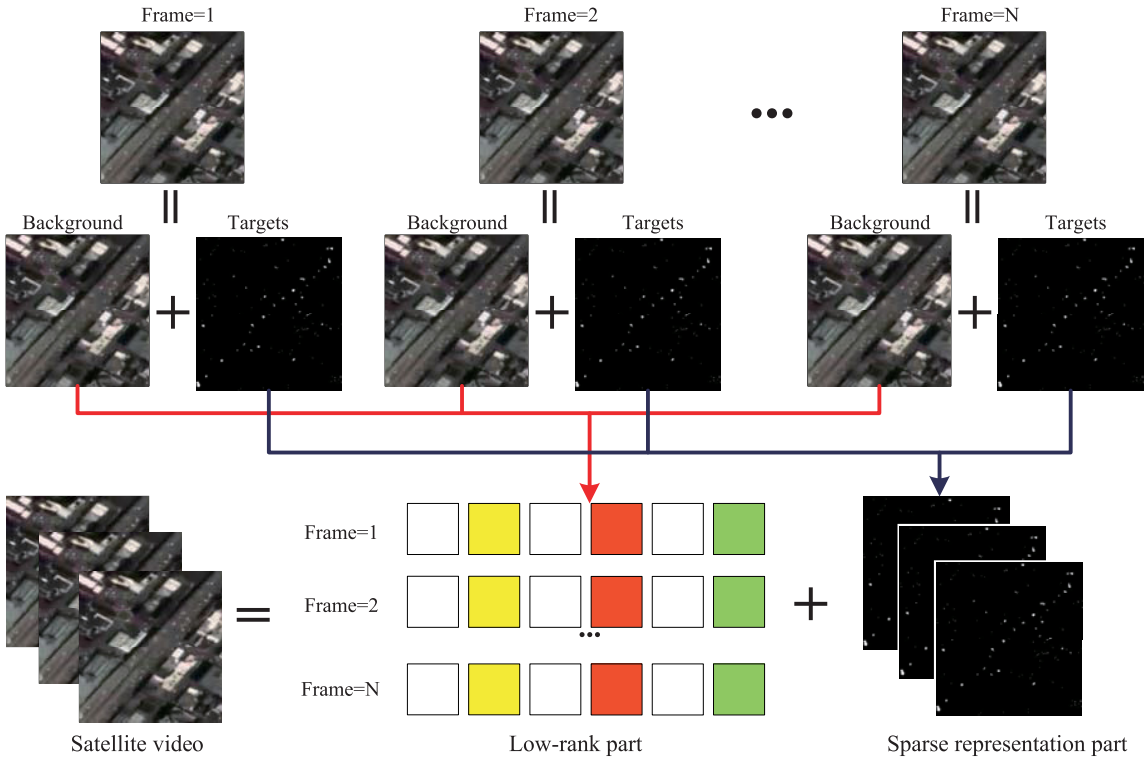


Fig. 1. Low-rank and sparse representation of a satellite video.

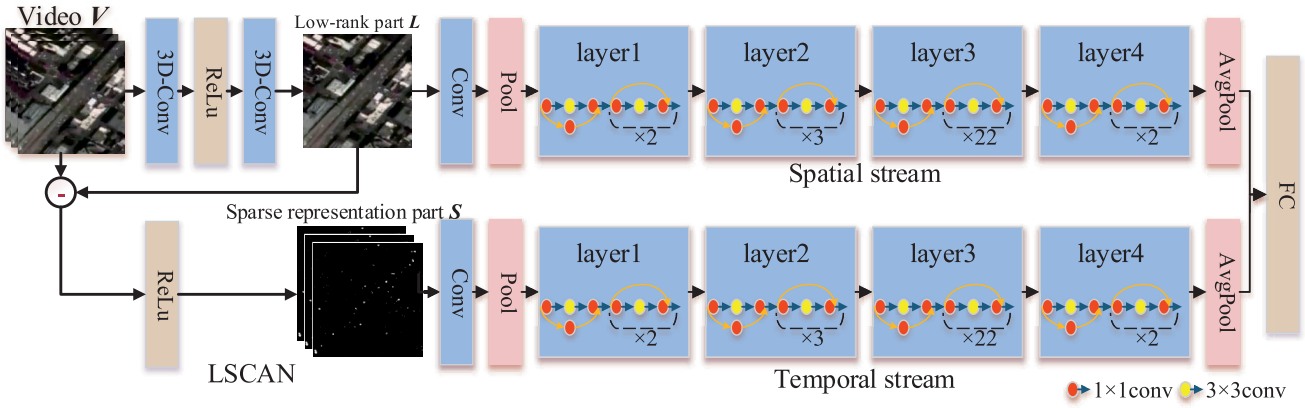


Fig. 2. Mainframe of LSRTN.

combining a label loss with a low-rank sparse representation loss gives the LSRTN the ability to fit the distribution of labels.

The remainder of this article is organized as follows. The method is presented in Section II. The data description and the experimental results are given in Section III. Section IV concludes this entire article.

II. Proposed Method

A. Mainframe

In this section, the LSRTN that possesses the benefits of both handcrafted and deep-learned features is described. The goal of the proposed work is to describe the video block that we call video scenes V by classifying the sequence of frames. Suppose that the spatial size of a satellite video scene V is $h \times \omega$, and the total number of frames is n. The main framework is shown in Fig. 2. The LSCAN is designed to separate V into the low-rank part L and sparse part S. L is a static image that represents the background, and S represents the motion of targets in each frame. There are two benefits to using the LSCAN. One benefit is that the LSCAN adds prior information to make small moving targets clearer. The other benefit is that the LSCAN reduces the redundant information of the background using the low-rank prior.

A two-stream network is used to extract spatial features from L and temporal features from S. The network is composed of three parts, i.e., the temporal feature extraction

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stream, the spatial feature extraction stream, and the classifier. The spatial features and temporal features are merged to obtain the category of V using the classifier. L and S are used instead of a simple frame and optical flow. The low-rank sparse net enhances the features of the original video and makes two-stream networks more efficient for SVSC. Moreover, training a typical two-stream net needs to calculate the optical flow first, but the quality of the optical flow deeply influences the temporal features. Supervised feature extraction by merging the loss in the training step instead of the optical flow can solve this problem.

B. Low-Rank Sparse Component Analysis Network

We observed that the most influential factor for the robustness of engineered features is the distribution of training data. Nonuniform sampling requires more data to balance deep learning networks. Adding prior information to change the space is a common thought, i.e., using unsupervised clustering before classification. It is useful yet laborious to select a proper prior method with appropriate parameters to reorganize the data. Naturally, combining the prior method to reorganize the data with the label information to adjust the parameters can be a reasonable method, i.e., obvious priors should be used properly to solve actual questions. This section introduces our proposed LSCAN to achieve this. As shown in Fig. 2, the helpful knowledge to solve the small target question is that a satellite scene can be divided into a static background image, moving targets, and noise

$$V = L + S + N. \tag{1}$$

Suppose that V is the sequence of frames of the input video scene, where $V \in R^{W\times H\times N}$; and L is the sequence of background images. S is the sequence of images of moving targets, and N is the noise. We can assume that the backgrounds of each frame are approximately the same as the shooting area of the satellite moving slowly. Therefore, the sequence of background images L can be regarded as low rank in the time dimension. Most moving targets in satellite video scenes are tiny, which means that an image in $S + N$ is sparse.

The proposed method first uses two 3-D-Conv layers to obtain the low-rank part L from the input satellite video scene V. Then, $S + N$ is smoothed by an ReLU layer to obtain sparse part S. Suppose that the size of V is $w \times h \times n$, where $w \times h$ is the frame size, and n is the number of frames. The first 3-D-Conv layer aims to smooth the frame. The size of first the 3-D-Conv layer is set to $c \times 3 \times 3 \times 3$, where c is the number of the 3-D-Conv kernels, and padding is [1, 1, 0]. The output of the first 3-D-Conv is $c \times w \times h \times n - 2$. Then, we use a second 3-D-Conv layer to reduce the dimension. The size of the second 3-D-Conv layer is set to $r \times 3 \times 3 \times (n - 2)$, and the padding is [1, 1, 0]. The output of the second 3-D-Conv layer is $r \times w \times h \times 1$, where r is the rank that we want. That is, setting the parameters of 3-D-Conv can ensure the rank of L. If rank $= 1$, a static image represents the background as multiframe sparse images representing the foreground are obtained.

$V - L$ is the sum of the moving targets and noise. In satellite video, $S + N$ is sparse at the pixel level. The signal of S is stronger than the noise N, so we use an ReLU layer to filter the noise. ReLU function can sparse out the negative value, that is to say, the sparsity of the ReLU layer's output is adjustable by the backward [33]. When the ReLU layer is used as a sparse filter, it is a threshold filtering. The feedback function and the features can adjust the threshold filtering by the training step to ensure the results. The mathematic expectation of the noise is always less than the targets, so the ReLU layer can filter part of the noise and ensure the sparsity of the output. Using each image of L, we calculate $V - L$ by using each frame of V, so the size of $S + N$ is $r \times w \times h \times n$. In the training step, we rewrite the loss function to restrict L to be low rank and S to be sparse using backward propagation and labels.

C. Low-Rank Sparse Representation Two-Stream Network

In this section, we introduce the proposed two-stream approach in detail. The approach consists of two streams: a spatial feature extraction stream and a temporal feature extraction stream, as shown in Fig. 2. Input video V is first decomposed into the low-rank part L and the sparse representation part S. Then, the model extracts static features and motion features from L and S using a spatial stream and a temporal stream [34]–[36]. Finally, the classifier that optimizes the static and motion features by fusing the features is proposed. We stack the features of the two streams and use a fully connected layer to make the final classification.

Both streams adopt ResNet-100 [31] as the base network. The spatial stream takes only one static image as the input, while the temporal stack images contain moving targets as input. The weights pretrained on ImageNet [37] are used for the streams.

The output features of the spatial stream are f_s and the output features of the temporal stream are f_t. We concatenate the spatial features f_s and temporal features f_t as the output

$$f = f_s \oplus f_t. \tag{2}$$

The spatial–temporal features f and a fully connection layer are used to classify the satellite video V. The cross-entropy loss is defined by

$$\mathcal{L}_1 = -\sum_{i=1}^{K} y_i \log(p_i) \tag{3}$$

where K is the number of categories, y_i is the labels, and p_i is the output of the two-stream network.

D. Training Strategy

The LSCAN can be an unsupervised learning network, but training the network is slow. Moreover, it is unnecessary to advance a general model to separate static information and motion information. It is sufficient to use the label information that is appropriate for the dataset to adjust the low-rank sparse network. The goal of the proposed method is

$$\min\ (\mathcal{L}_1), \quad \text{s.t. } V = L + S. \tag{4}$$

L should be restricted to low rank, and S should be sparse. The noise N is filtered by the ReLU layer that we can ignore.

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TABLE I
NUMBER EACH CATEGORY OF SVSC DATASET

	bridge	freeway	intersection	overpass	parkinglot	round	runway	water-area	habor
Busy	12	11	48	24	182	7	45	38	179
Midbusy	0	122	0	0	0	0	0	0	0
Free	76	598	344	164	224	48	218	26	0

L can be restricted to a low rank by the convolution layers of the LSCAN. We can restrict the sparsity of S using the L0-norm. The L0-norm is the sparsity constraint. We use the L1-norm to restrict S instead of the L0-norm. The L1-norm is approximately equal to the L0-norm, but it is smoother and makes it easier to calculate the gradient. Minimizing the L1-norm will make S sparse. The L1-norm is added to the loss function so that the backward propagation can adjust the weights of the LSCAN by the label information. We can rewrite the loss function and the goal

$$\mathcal{L}_2 = \mathcal{L}_1 + \lambda \times \|S\|_1 \tag{5}$$

and

$$\min\ (\mathcal{L}_2), \quad \text{s.t. } V = L + S. \tag{6}$$

There is a situation in which l1loss $= 0$, so we need another constraint to ensure that S is not too sparse. It was found that, when l1loss decreased, $V - L - S$ changed distinctly. We use l2loss to ensure that S not too sparse. Considering this, we can also add this constraint to the loss function and rewrite the goal

$$\mathcal{L} = \mathcal{L}_1 + \lambda \times \|S\|_1 + \mu \times \|V - S - L\|_2 \tag{7}$$

$$\min(\mathcal{L}). \tag{8}$$

Therefore, the LSCAN and the two-stream net are trained together using the loss function.

III. EXPERIMENTS

In this section, first, the SVSC dataset is introduced. Then, the effects of the LSCAN, the performances of the baseline methods, the comparison of different backbones of a two-stream network, and the proposed method on the SVSC dataset are given.

A. Dataset

We evaluated the proposed methods and performed ablation studies on the SVSC datasets. We divided seven satellite videos into video scenes and chose 18 categories of moving scenes extracted by the EOI method to create a new dataset for the experiments. All the video scenes are extracted from nine Jilin-1 satellite videos and the previously published videos of the Jilin-1 satellite. The width of a satellite video is 12 km, and the height is 5 km; the spatial resolution is 1 m, and the frame rate is 30. A small scene has approximately 325 frames, and a frame has 256 $\times$ 256 pixels. We focus on statuses such as busy and free of the scenes for the applications. The categories refer to the application of traffic and shipping. Table I gives the number of each category of the SVSC dataset. The 18 categories are the following: Busy-Bridge (BB), Busy-Freeway (BF), Busy-Intersection (BI), Busy-Overpass (BO), Busy-Parking-lot (BP), Busy-Round (BR), Busy-Runway (BRW), Free-Bridge (BB), Free-Freeway (BF), Free-Intersection (BI), Free-Overpass (BO), Free-Parking-lot (BP), Free-Round (BR), Free-Runway (BRW), Busy-Harbor (BH), Medium-Busy-Freeway (MBF), Busy-Water-area (BW), and Free-Water-area (FW). Fig. 3 shows the examples of the dataset. In this dataset, we only consider the status of moving scenes. Static scenes are excluded from the dataset because discriminating the motion statuses is more meaningful for satellite videos.

B. Experimental Settings

We use the confusion matrix, overall accuracy (OA), and average accuracy (AA) to evaluate the methods. $C_{i,j}$ of confusion matrix C can be defined as the number of test samples that are category i but predicted to be category j. $C_{i,j}$ is generally used to evaluate the classification accuracy. The normalized confusion matrix calculates $C_{i,j}$ for category i. OA is the true positive rate of the entire sample, and AA calculates the AA of all the categories. The kappa coefficient is also useful when the samples are unbalanced, but it is approximately equal to the AA in most cases. We choose the AA because it is more convenient to calculate and more obvious to compare. 797 training samples and 1569 testing samples are used in the experiments. We set lr $= 0.001$ and the batch size $= 32$ to train the networks. In the proposed method, we set the hyperparameters $\mu = 0.0001$ because the L2-norm is much larger than the L1-norm and $\lambda = 0.5$.

C. Comparison

In this section, we implement the baseline methods. The 3-D-CNNs, LSTMs, and two-stream networks are the state-of-the-art and typical computer version methods. These video classification methods achieved better performances on UCF101 [1], Kinetics [44], HDMB51 [45], and so on. The most important work is proving the effectiveness of the LSCAN. We will discuss the results of the low-rank part, the sparse representation part, and the experimental results in the following.

1) Low-Rank Part: Generally, the first frame or some shots of a video are commonly used to extract the spatial features for two-stream networks. It is easy for those methods to obtain the spatial features from one frame, but considering all the frames is helpful for the spatial feature extraction of complex backgrounds. Therefore, strengthening the background features of the target of interest, such as those of

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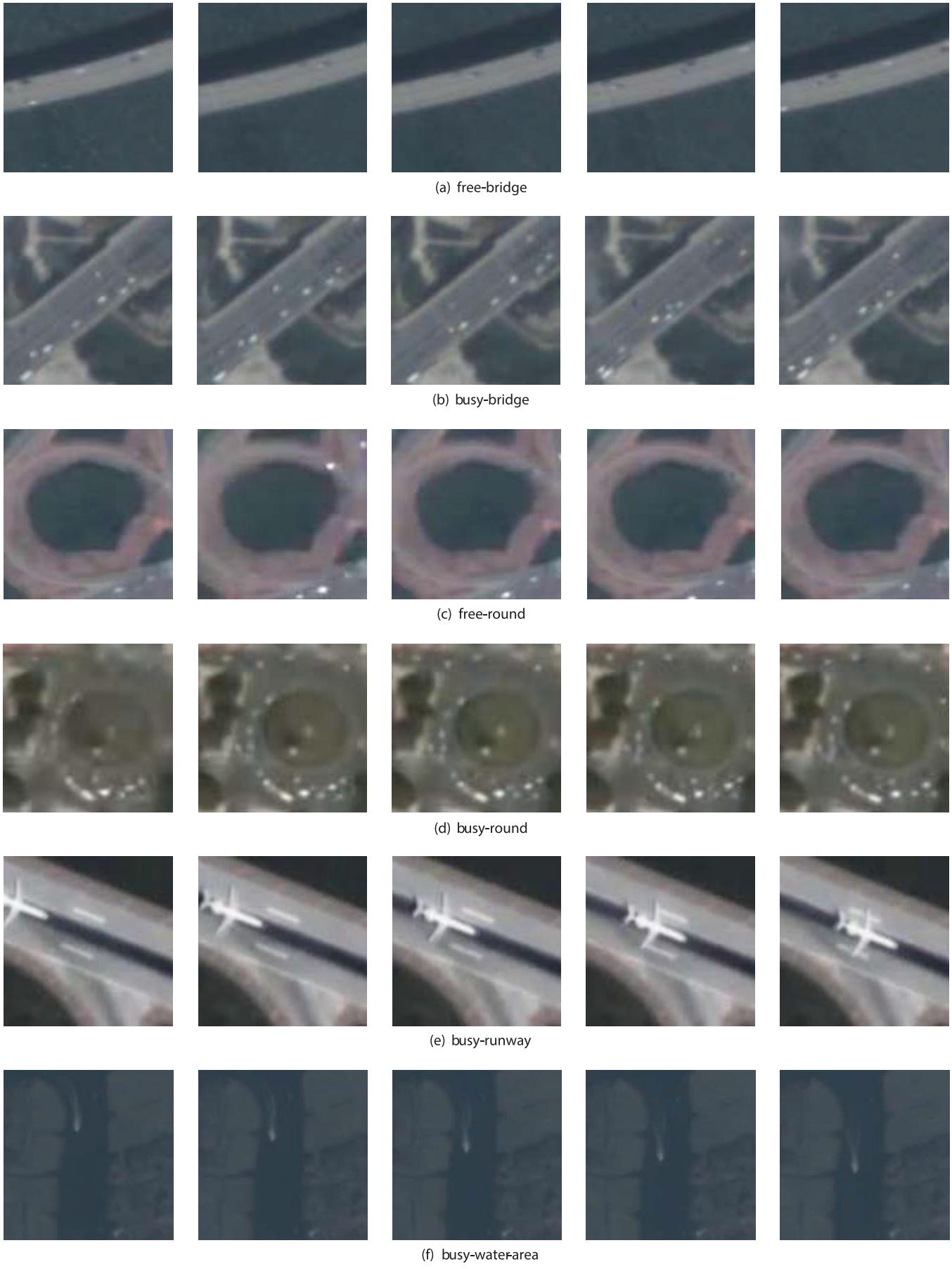


Fig. 3. Examples of the dataset. (a) Free-bridge. (b) Busy-bridge. (c) Free-round. (d) Busy-round. (e) Busy-runway. (f) Busy-water-area.

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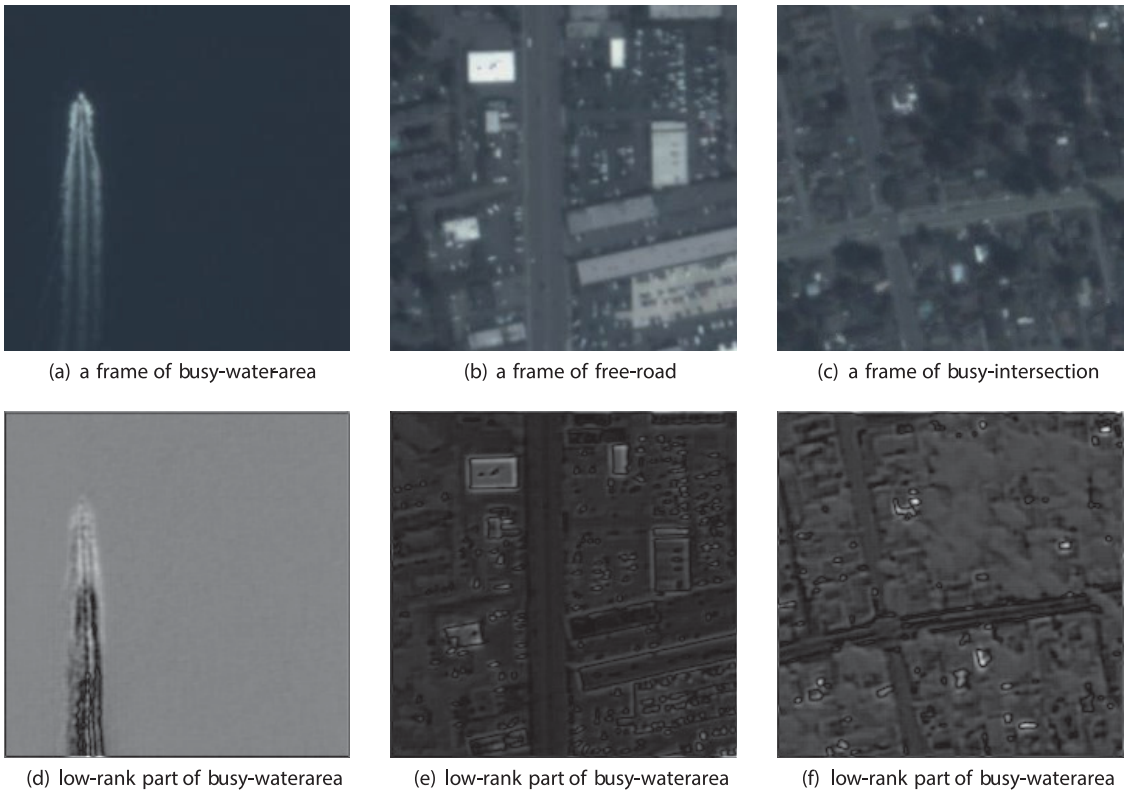


Fig. 4. Low-rank part results of LSCAN. (a) Frame of busy-water-area. (b) Frame of free-road. (c) Frame of busy-intersection. (d) Low-rank part of busy-water-area. (e) Low-rank part of busy-water-area. (f) Low-rank part of busy-water-area.

TABLE II
RESULT OF COMPARED METHODS

	BB	**BF**	**BI**	**BO**	**BP**	**BR**	**BRW**	**FB**	**FF**	**FI**	**FO**	**FP**	**FR**	**FRW**	**BH**	**MBF**	**BW**	**FW**	**OA**	**AA**
CNN	0	0.143	0	0	0.057	0	0.103	0.360	0.545	0.040	0.046	0.041	0.062	0.243	0.689	0	0.250	0.625	0.256	0.178
C3D [10]	0	0.143	0	0	0.057	0	0.103	0.360	0.545	0.040	0.046	0.041	0.062	0.243	0.689	0	0.250	0.625	0.468	0.290
I3D [8]	0.250	0.857	0.111	0.188	0.262	0.500	0.517	0.520	0.862	0.662	0.556	0.737	0.375	0.944	0.857	0.738	**0.958**	0.688	0.699	0.588
CSN [38]	0	0.857	0.028	0.062	0	0	**0.793**	0.140	0.756	0.851	0.343	**0.858**	0	0.944	0.723	0.363	0.750	0	0.616	0.415
R2plus1D [39]	0.125	0.857	0.083	0	0.229	0.250	0.276	0.460	0.731	0.583	0.472	0.635	0.312	0.868	0.832	0.725	0.833	0.625	0.612	0.494
CNN+LSTM [13]	0	0	0	0	0.525	0	0	0.080	0.857	0.737	0.250	0.527	0.125	0.868	0.857	0.312	0	0.625	0.602	0.320
LRCN [40]	0	0	0	0	0.516	0	0	0	0.814	0.316	0.176	0.297	0	0.583	0.605	0.150	0.458	0	0.447	0.218
CNN+MLP	0	0.857	0	0	0.393	0	0.276	0.260	0.839	0.500	0.528	0.635	0.750	0.826	0.815	0.562	0.542	0.625	0.626	0.467
two-stream [14]	0.125	0.857	0.083	0.063	0.328	0	0	0.655	0.480	**0.930**	**0.798**	0.694	0.757	0.781	**0.924**	0.725	**0.958**	0.688	0.760	0.598
SlowFast [41]	0.125	0.857	0.111	0.125	0.213	**0.750**	0.759	0.620	0.776	0.632	0.528	0.554	0.344	**0.958**	0.916	0.650	0.917	0.688	0.656	0.585
SlowOnly [41]	0.250	0.857	0.028	0	0.344	0.500	0.310	0.540	0.847	0.632	0.518	0.574	0.219	0.889	0.916	0.613	0.875	0.688	0.660	0.533
TPN [42]	0.125	0.857	0.139	0.188	0.221	0	0.345	0.480	0.857	0.654	0.472	0.662	0.281	**0.958**	0.857	0.738	0.875	0.688	0.672	0.522
TSM [43]	0.250	0.857	0.028	0.250	0.328	0.500	0.621	0.420	0.789	0.741	0.574	0.696	0.531	0.889	0.874	0.700	0.875	0.688	0.688	0.589
TSN [16]	**0.375**	0.857	0.083	0.062	0.287	0.250	0.517	0.440	0.864	0.785	0.620	0.743	0.594	0.917	0.891	0.725	**0.958**	0.688	0.723	0.592
LSRTN	**0.375**	**1.00**	**0.361**	**0.625**	**0.746**	0.250	0.655	**0.680**	**0.882**	0.886	0.769	0.716	**0.969**	0.938	0.899	**0.825**	**0.958**	**0.813**	**0.812**	**0.742**

a road or runway, is useful for classification. Fig. 4 shows the result of the low-rank part extracted by the LSCAN. Fig. 4(a)–(c) shows frames extracted from videos of typical scenes. Fig. 4(d)–(f) gives the low-rank part that is extracted by LSCAN. The figures show that the low-rank parts can strengthen the backgrounds.

2) Sparse Representation Part: Extracting appropriate temporal features is important for SVSC. Unsupervised methods can provide more information to temporal features as the optical flow is the most commonly used method for motion information preextraction. The optical flow was applied in two-stream-based methods and obtained better performances in common nature videos, but it failed in satellite videos. The temporal features of massive numbers of small targets cannot be obtained from one frame or a few frames, and it is difficult to obtain the features from the entire frame

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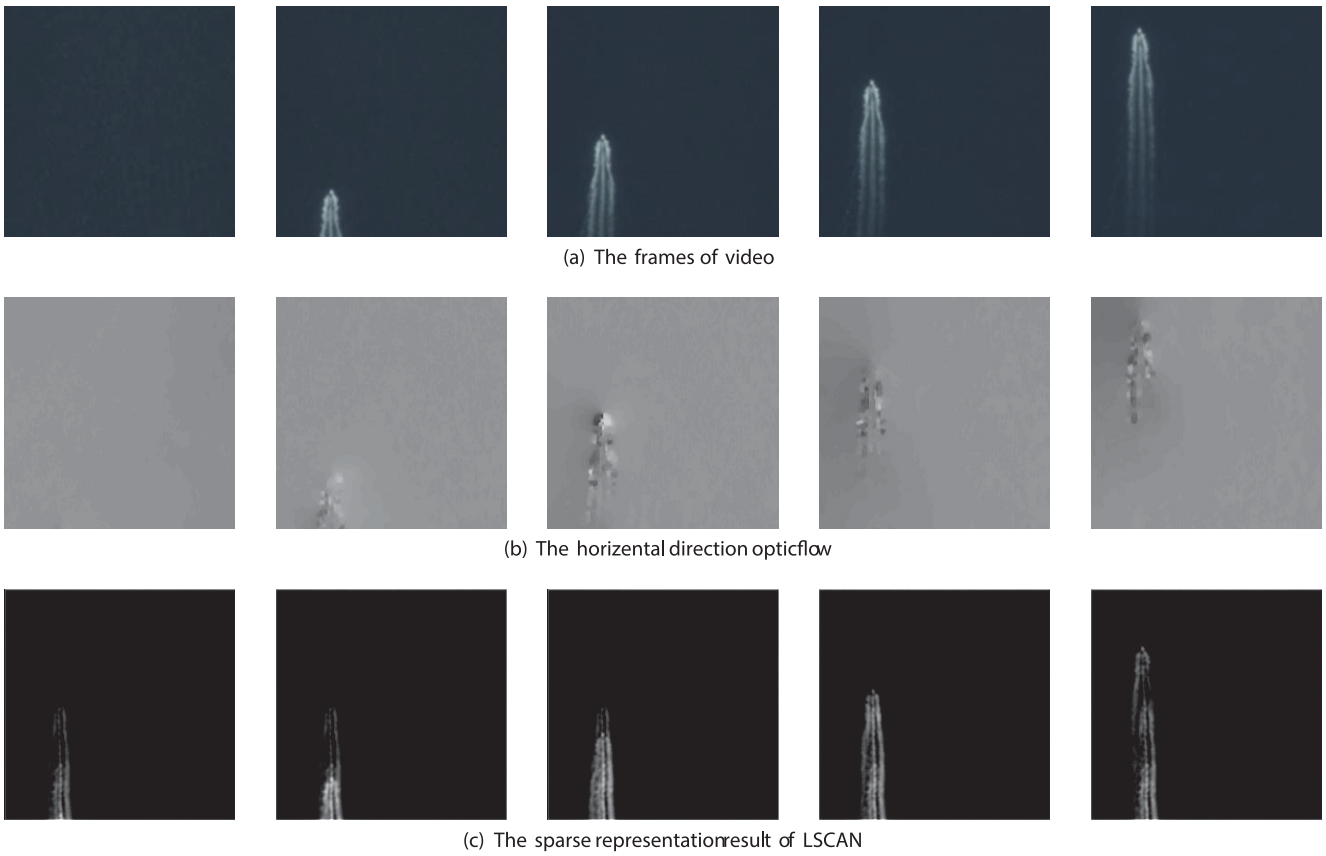


Fig. 5. Results of optical flow and LSCAN of the busy water-area scene. (a) Frames of video. (b) Horizontal direction optical flow. (c) Sparse representation result of LSCAN.

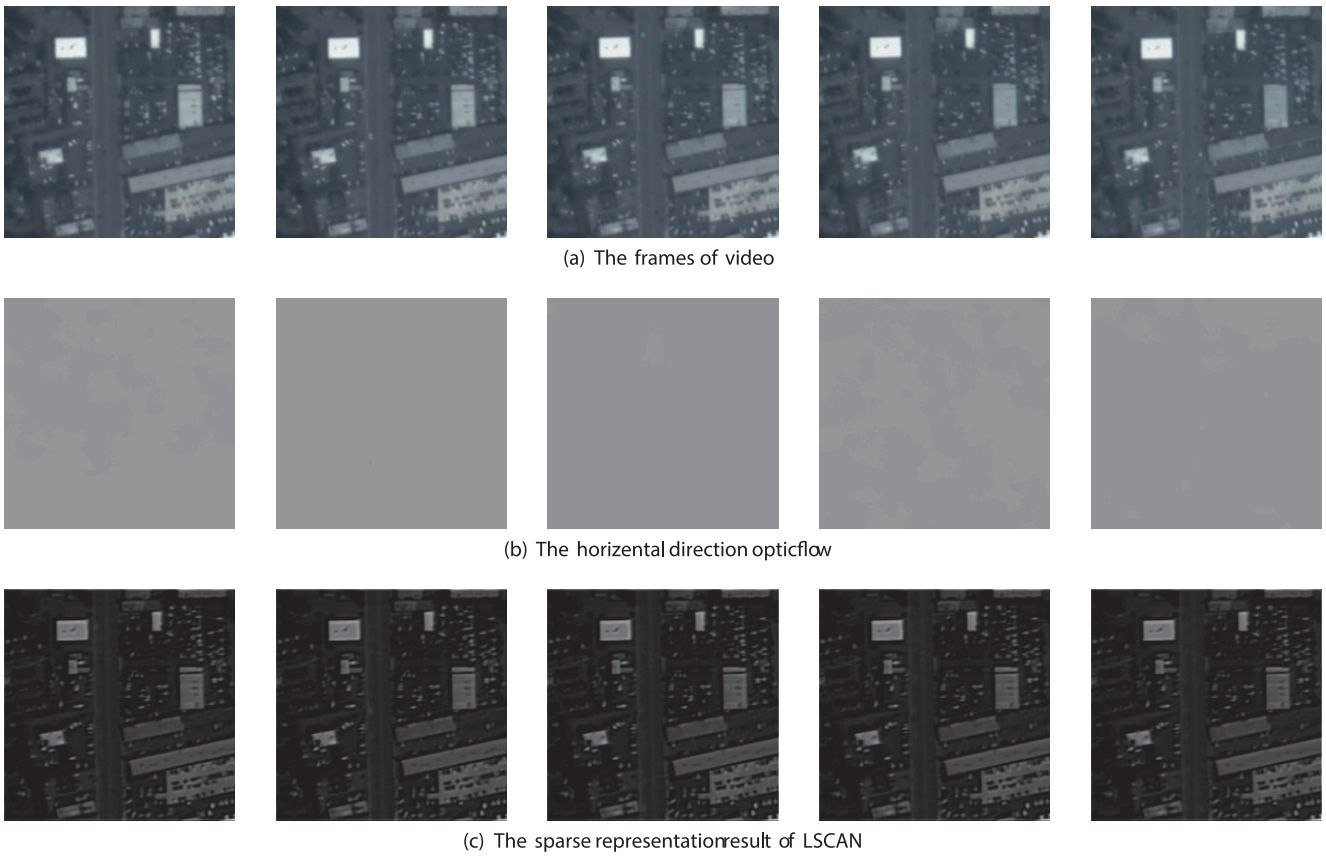


Fig. 6. Results of optical flow and LSCAN of the free-road scene. (a) Frames of video. (b) Horizontal direction optical flow. (c) Sparse representation result of LSCAN.

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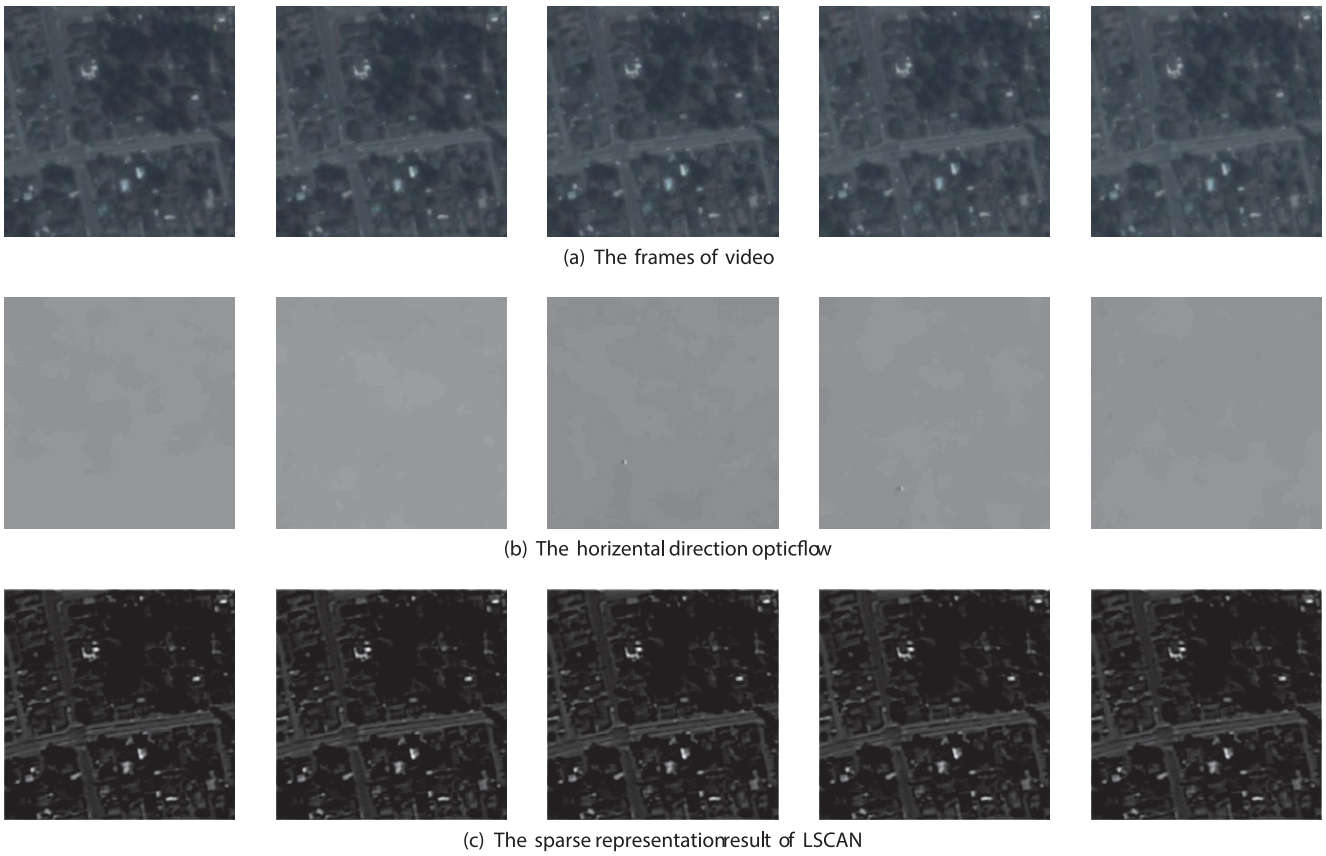


Fig. 7. Results of optical flow and LSCAN of the BI scene. (a) Frames of video. (b) Horizontal direction optical flow. (c) Sparse representation result of LSCAN.

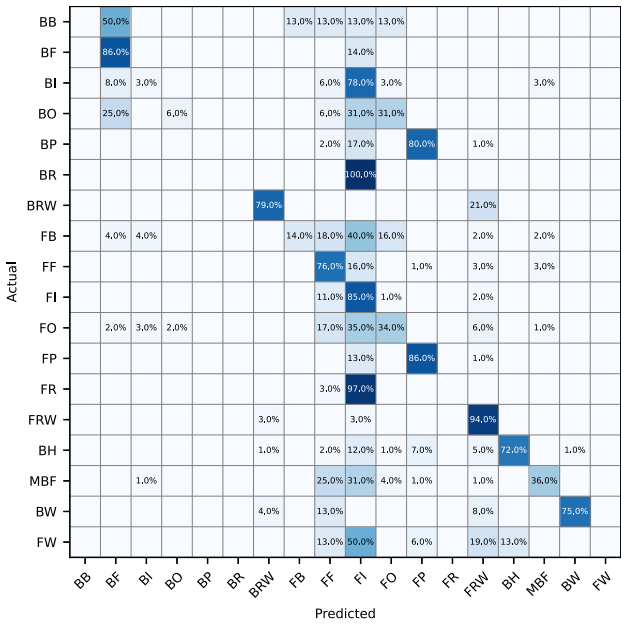


Fig. 8. Normalized confusion matrix of CSN.

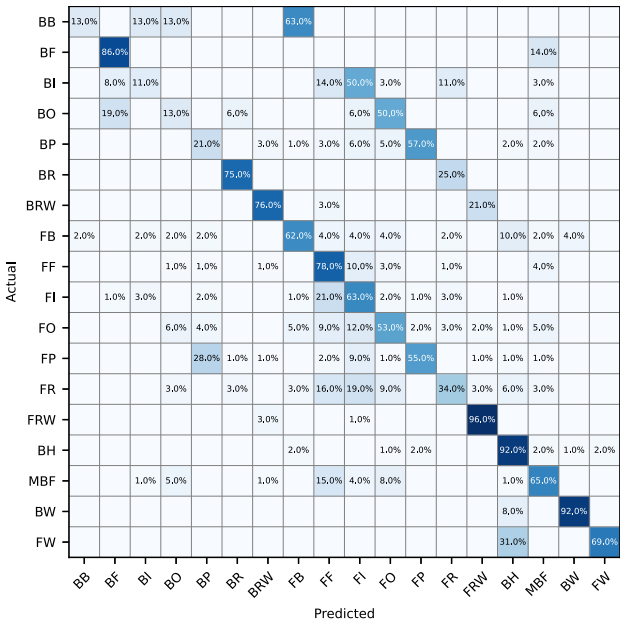


Fig. 9. Normalized confusion matrix of slow–fast networks.

due to jittering, tiny changes, and noises. To illustrate this, we compared the results of the optical flow with the sparse representation parts of the LSCAN. Fig. 5 shows the results in a BW scene. The optical flow results were obtained by the state-of-the-art method, FlowNet. The optical flow obtained satisfactory results in scenes with bigger larger and simple backgrounds. Both the optical flow and LSCAN could extract the motion of ships. Fig. 6 gives the results for a free city scene. We can observe that the optical flow could not obtain anything as the LSCAN ignored the road without any moving

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TABLE III
RESULT OF DIFFERENT BACKBONE

	BB	BF	BI	BO	BP	BR	BRW	FB	FF	FI	FO	FP	FR	FRW	BH	MBF	BW	FW	OA	AA
two-stream+ResNet101	0.125	0.857	0.083	0.063	0.328	0	0	0.655	0.480	0.930	0.798	0.694	0.757	0.781	0.924	0.725	0.958	0.688	0.760	0.598
two-stream+ResNet18	0.250	0.857	0.028	0	0.451	0.500	0.414	0.340	0.859	0.658	0.472	0.655	0.438	0.854	0.697	0.513	0.667	0.667	0.645	0.518
two-stream+Vgg19	0	0.857	0	0	0.50	0	0.276	0.30	0.882	0.789	0.574	0.608	0.563	0.931	0.924	0.50	0.668	0	0.695	0.465
two-stream+vgg16	0	0.857	0	0	0.532	0.000	0.138	0.260	0.862	0.772	0.602	0.561	0	0.958	0.916	0.5	0.708	0	0.675	0.426
EfficientNet	0	0.857	0.139	0.188	0.276	0	0.30	0.480	0.857	0.654	0.472	0.662	0.281	0.931	0.857	0.738	0.875	0.688	0.668	0.514
InceptNet	0	0.857	0.111	0.188	0.262	0.500	0.517	0.520	0.862	0.662	0.556	0.737	0.375	0.944	0.857	0.738	0.958	0.688	0.698	0.574
LSRTN-lossfunction	0	0.428	0	0.063	0.328	0.25	0	0.655	0.480	0.930	0.796	0.694	0.757	0.781	0.924	0.763	0.958	0.5	0.665	0.530

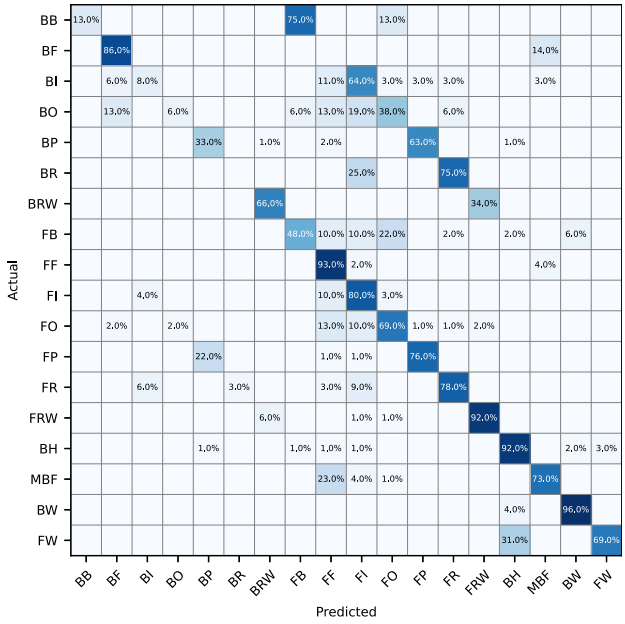


Fig. 10. Normalized confusion matrix of two-stream networks.

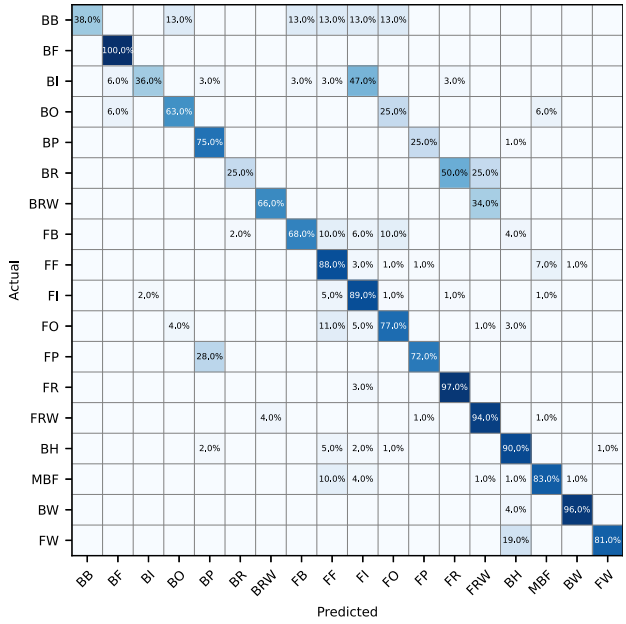


Fig. 11. Normalized confusion matrix of LSRTN.

target. Fig. 7 shows the results in BI. Optical flow could only obtain the motion of a larger car moving from north to south, but LSCAN obtained all the targets in the road of each frame. The features of the free north–south road are different from those of the busy east-west road. The comparison shows that LSCAN can obtain better features of scenes with a low signal-to-noise ratio (SNR) by sparse representation.

3) Experimental Results: The precision of each category, OA, and AA are given in Table II. The table shows that the results of CNN-LSTM-based methods are worse than those of other methods. These methods have high demands for the quality of the spatial features extracted by CNNs. Unfortunately, it is difficult for a CNN to extract the feature of small targets from static images because all the moving statuses are mistaken and classified as free. More improvement of the spatial feature extraction by combining multiframes is needed. 3-D-CNN-based methods obtained better results than the CNN-LSTM-based methods, but 3-D-CNN-based methods also relied on the parameters of convolutional layers. The normalized confusion matrix of the CSN is given in Fig. 8. It is difficult to select the parameters of convolutional layers, especially for dim small targets. Extracting the spatial features of small targets requires small convolutional masks. However, extracting temporal features of moving targets requires large spatial and long-term convolutional masks. This contradiction makes classification difficult to achieve. Even in 2-D image classification areas, adjusting the parameters of convolutional masks is very hard. The other question regarding the 3-D-CNN is that the number of training samples cannot be satisfied because of the dim explosion. Transferring the parameters of the convolutional mask can solve some questions, but this is very difficult to achieve for deep learning.

Figs. 9 and 10 show the confusion matrix of slow–fast networks and two-stream networks. These figures show that the two-stream-based methods with tiny convolution kernels are more suitable for SVSC. The confusion matrices illustrate that it is difficult to determine the moving status of scenes in most methods. Those two-stream-based methods obtained better results due to how the original optical flow worked, but, in some scenes, especially in low SNR scenes, even the optical flow cannot obtain better motion features. In particular, the TSN obtained a worse result than the two-stream convolutional networks. Lacking training samples of small target scenes made the result unsatisfactory. Fig. 11 shows the

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normalized confusion matrix of the LSRTN. The LSRTN that benefits from low-rank and sparse representation can better classify scenes that contain massive numbers of dim small targets.

In Table III, we compare different backbones of two-stream networks, the EfficientNet, and the InceptNet to illustrate that the LSCAN is working. The results show that different layers and parameters cannot raise the classification accuracy. We subtracted the sparse constrained loss function of the LSRTN and got a worse result than the two-stream networks because it is hard to have ample training. Those results illustrate the effectiveness of the proposed network.

IV. Conclusion

In this article, we proposed the LSRTN method for SVSC. By comprehensively analyzing the challenges of SVSC, the LSRTN, which can efficiently and effectively extract the spatial features and temporal features from satellite videos, was designed. The method is preferred for classifying low SNR satellite video scenes that contain the motions of dim small targets. The experiments illustrated that the proposed method extracts the motion information better than the baseline methods. Moreover, it is much faster than the methods using optical flow. The experiments also demonstrated the performance of the LSRTN.

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